

C++ Programming Tasks - Code and Output

TASK 1: CGPA Calculator (C++ Code)

```
#include
#include
using namespace std;

int main() {
    int numCourses;
    cout << "Enter the number of courses: ";
    cin >> numCourses;

    double grade, creditHours;
    double totalGradePoints = 0.0, totalCredits = 0.0;

    cout << "\n--- Enter Grade and Credit Hours for each course ---\n";

    for (int i = 1; i <= numCourses; i++) {
        cout << "\nCourse " << i << ": \n";
        cout << "Grade: ";
        cin >> grade;
        cout << "Credit Hours: ";
        cin >> creditHours;

        totalGradePoints += grade * creditHours;
        totalCredits += creditHours;
    }

    double cgpa = totalGradePoints / totalCredits;

    cout << fixed << setprecision(2);
    cout << "\n===== \n";
    cout << "Total Credits : " << totalCredits << endl;
    cout << "Total Grade Points : " << totalGradePoints << endl;
    cout << "Final CGPA : " << cgpa << endl;
    cout << "===== \n";

    return 0;
}
```

Sample Output:

```
Enter the number of courses: 3
Course 1 -> Grade: 9 Credit: 4
Course 2 -> Grade: 8 Credit: 3
Course 3 -> Grade: 7 Credit: 2
```

Total Credits: 9

Total Grade Points: 73
Final CGPA: 8.11

TASK 2: Login & Registration System (C++ Code)

```
#include
#include
#include
using namespace std;

void registerUser() {
    string username, password;
    cout << "----- User Registration -----\\n";
    cout << "Enter Username: ";
    cin >> username;
    cout << "Enter Password: ";
    cin >> password;

    string filename = username + ".txt";
    ofstream file(filename.c_str());

    if (file.is_open()) {
        file << username << endl << password;
        file.close();
        cout << "Registration Successful!\\n";
    } else {
        cout << "Error Creating User File.\\n";
    }
}

void loginUser() {
    string username, password, u, p;
    cout << "----- User Login -----\\n";
    cout << "Enter Username: ";
    cin >> username;
    cout << "Enter Password: ";
    cin >> password;

    string filename = username + ".txt";
    ifstream file(filename.c_str());

    if (file.is_open()) {
        getline(file, u);
        getline(file, p);
        file.close();

        if (u == username && p == password) {
            cout << "Login Successful! Welcome, " << username << "!\\n";
        } else {
            cout << "Invalid Username or Password!\\n";
        }
    }
}
```

```

}
} else {
cout << "User not found. Please register first.\n";
}
}

int main() {
int choice;
cout << "==== Login & Registration System =====\n";
cout << "1. Register\n";
cout << "2. Login\n";
cout << "Enter your choice: ";
cin >> choice;

switch (choice) {
case 1:
registerUser();
break;
case 2:
loginUser();
break;
default:
cout << "Invalid Choice!\n";
break;
}

return 0;
}

```

Sample Output:

```

==== Login & Registration System =====
1. Register
2. Login
Enter your choice: 1
Enter Username: user1
Enter Password: pass123
Registration Successful!

==== Login & Registration System =====
Enter your choice: 2
Enter Username: user1
Enter Password: pass123
Login Successful! Welcome, user1!

```